

Grades for Erneste Ntezirizaza

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Course

Arrange By

Mission Capstone



Due date



Apply

Name	Due	Submitted	Status	Score	
Honour code Assignments	25 May by 23:59	12 May at 9:34		1 / 1	
Initial software product/solution demonstration Assignment 2: Initial Design and Development [F:A]	4 Jun by 23:59		missing	0 / 0	
Assignment 1: Research Proposal Assignment 1: Research proposal [F:A]	9 Jun by 23:59	9 Jun at 23:39		15.6 / 19	  
Final version of the product/solution Assignment 3: Final Capstone Implementation [S:A]	8 Jul by 23:59	8 Jul at 23:41		10 / 15	 

Assessment by Dirac Murairi

[Close Rubric](#)

3. Implementation and Testing

Criteria	Ratings			Points
Testing results & strategies	4 to >2 pts Excellent Multiple relevant testing strategies clearly demonstrated; strong evidence of	2 to >1 pts Good Good testing evidence with relevant strategies and some variation in conditions; no	1 to >0 pts No marks No meaningful testing evidence provided.	1.5 / 4 pts

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3. Implementation and Testing

Criteria	Ratings			Points
	different inputs, edge cases, and performance across hardware/software environments. evaluation across different hardware/software environments. Comments "testing" here means exercising UI controls to confirm a state changes, not stress-testing the system under conditions it wasn't told to expect. Genuine boundary/edge-case testing for this kind of system would look like: running an episode to battery exhaustion without operator intervention and observing what the policy actually does, spawning obstacles at a randomized rate baked into the environment rather than a button press, running dozens of episodes across all nine zones and checking whether terrain/seed/reward numbers actually vary in ways traceable to real geospatial differences, and deliberately breaking the telemetry pipeline (bad zone ID, dropped connection) to see how the system degrades. None of that is present, and what is present doesn't consistently match what's claimed.			
Analysis of results	2 to >1.5 pts Excellent Detailed analysis of how results were achieved or how they missed the proposal objectives; strong linkage to project scope.	1.5 to >0 pts Good Report lists only findings without deeper analysis.	0 pts No marks No analysis of results.	1 / 2 pts
	The Analysis section is structured as a findings statement rather than an actual analysis. It asserts that the central objective "was achieved" and supports this with two numbers, 1,182 seeds placed and a 96.42 average reward across 12 episodes, but neither number is explained: there's no discussion of what a good or bad average reward looks like for this reward function, no comparison against any baseline, and no reasoning connecting the number to why the system worked. The one place the section does something closer to real analysis is the partially-met objective on reseed pipeline visibility, where it identifies a specific limitation (the visual replant only completes at the next full mission cycle) and explains why (a short demo window can't capture a full mission cycle). That's the right shape for this criterion, but it's the only instance of it in the whole section. The bigger issue is that the claims in this section don't hold up against the evidence actually available. The section states the system runs "against a live PPO+CNN policy, not scripted animations," and that all seven functionalities are demonstrated, but the screenshots show flat, undifferentiated terrain with no visible connection to real geospatial			

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	data, a battery reading stuck at 100% through an episode where seeds have already dropped and obstacles are active, and obstacles positioned directly on the drone rather than being routed around, none of which match the claims being analyzed. A results analysis is only as good as its grounding in what was actually observed, and right now the write-up analyzes a version of the system's behavior that doesn't match what the visual evidence shows. Before this can be scored higher, the analysis needs to be rebuilt around what the screenshots and any raw telemetry logs actually demonstrate, including naming the mismatches (terrain fidelity, obstacle behavior, battery drain) as findings in their own right rather than leaving them unaddressed while claiming the objective was fully met.			
	3 to >2 pts Excellent Clear, well-structured deployment plan with steps, tools, and environments fully documented; successfully deployed and verified through testing in the target environment.	2 to >1 pts Satisfactory Plan present but lacks clarity or completeness; system deployed but partially incomplete or difficult to reproduce; partial verification shown.	1 to >0 pts Poor No clear deployment plan; deployment not achieved or only minimally attempted; no verification provided.	3 / 3 pts
Functionality, scope alignment & algorithms	2 to >1 pts Excellent Core functionalities fully implemented and technically sound; strong alignment with approved proposal; effective use of algorithms and custom logic with thoughtful, efficient implementation.	1 to >0 pts Good Most functionalities implemented with minor issues; mostly aligned with proposal; good technical logic with minor limitations.	0 pts Poor Minimal or non-functional implementation; weak alignment with approved scope; minimal evidence of technical problem-solving.	1 / 2 pts
	Comments			

Name**Due****Submitted****Status****Score****3. Implementation and Testing**

Criteria	Ratings			Points
	The simulation DOES NOT REFLECT RWANDA FORESTS/ECOYSTEM The evidence in these screenshots doesn't support the level of implementation claimed in the README, and the gap matters because the whole proposal's justification rests on terrain-adaptive, ecologically grounded placement. All four screenshots show the same flat, uniform green plane with a grey backdrop — there's no visible elevation, slope, soil variation, or land cover difference anywhere in the scene, despite the system being described as trained on six real Rwanda geospatial layers including a DEM and soil classification. A terrain-adaptive agent operating on real elevation and soil data should produce a visually varied surface, not a single flat color; what's shown here looks like a placeholder plane rather than a rendering of any actual terrain data. The zone label ("Central Plateau East 3/9") is present as text, but nothing about the rendered environment reflects what that zone's real geography would look like. The obstacle behavior in images 3 and 4 raises a similar concern. The red spheres appear directly adjacent to or on top of the drone rather than being routed around, which doesn't match the "detects and routes around dynamically spawned hazards mid-flight" claim in the README; there's no visible flight-path deviation, just obstacles co-located with the drone. Battery also reads 100% in all four images, including the one with obstacles turned on and seed count already dropped from 500 to 444, which means either the drain mechanic isn't running during this session or isn't reflected in the HUD, contradicting the claimed solar/battery energy system.			
Code Quality	2 to >1 pts Excellent Strong application of OOP principles; modular, maintainable, reusable, and well-structured code with clear naming conventions.	1 to >0 pts Good Good code structure with minor excessive issues in modularity duplication; or maintainability; some evidence of structured code but inconsistent practices.	0 pts Poor Organisation; monolithic scripts; weak maintainability.	1.5 / 2 pts
	Comments Cryptic single/double-letter variable names: lo, hi (line 61/81-85), lc (land cover, 107-113, 185), m (mapping dict, 108), mask/dist/prox in compute_disturbance (130-134), gmax (194) — none are self-descriptive; a reader has to infer meaning from context instead of the name. Underscore-prefixed "throwaway" names that aren't actually throwaway: _parts, _wp, _e, _pd in compute_disturbance() (117-128) — the underscore convention normally signals "unused," but these are			

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	used throughout the function; it's misleading and inconsistent with the rest of the file. Magic numbers with no named constants: <code>encode_lc()</code>'s <code>m</code> dict (108-109) hardcodes land-cover class codes (10, 20, 30...) to opacity-like scores (0.3, 0.8, 0.9...) with zero explanation of what the codes or scores mean; <code>compute_soil()</code> hardcodes weights 0.40/0.25/0.20/0.15 (104) and the pH constant 0.65 (103) inline instead of named constants. No comment explaining the "why" on non-obvious math: <code>compute_soil</code>'s weighted formula, <code>compute_disturbance</code>'s $1.0 / (1.0 + \text{dist} / 20.0)$ decay constant (133), and <code>compute_obstacle</code>'s 0.7/0.3 blend (150) all encode domain judgment calls with no explanation of where the numbers came from or why those weights. Inconsistent naming style within the same file: mixes verbose descriptive names (<code>elev_min</code>, <code>rain_stack</code>, <code>no_plant</code>) with terse/cryptic ones (<code>lo</code>, <code>hi</code>, <code>lc</code>, <code>m</code>) — no consistent convention for what deserves a full word vs. an abbreviation. <code>run()</code> is a 90-line procedural script, not a function — 9 sequential numbered print-annotated steps with no decomposition into sub-steps/helper functions; each step mixes I/O (<code>np.save</code> (http://np.save)), computation, and logging together instead of separating concerns (this is the "monolithic script" pattern the rubric explicitly penalizes). Reliance on <code>print()</code> for pipeline status instead of logging, and progress tracking via manual "[1/9]" string literals rather than a loop/config-driven step list — brittle if steps are added/removed later. Local imports scattered inside functions (from <code>rasterio.features</code> import <code>rasterize</code>, from <code>shapely.geometry</code> import <code>mapping</code>, import <code>pandas</code> as <code>_pd</code> inside <code>compute_disturbance()</code>, from <code>scipy.ndimage</code> import <code>uniform_filter</code> inside <code>compute_obstacle()</code>) instead of at module top — inconsistent with the top-level imports already present and no explanation for why these are deferred. No docstrings on most functions (<code>resample</code>, <code>norm</code>, <code>sanitise</code>, <code>encode_lc</code>, <code>compute_disturbance</code>, <code>run</code> — only <code>compute_slope</code> and <code>compute_obstacle</code> have them), despite this being the most numerically dense, least self-explanatory file in the ML codebase — exactly where comments explaining why (not what) are most needed.					
Technical report, repository & video	2 to >1 pts Excellent Well-structured, professional report with clear technical explanations; professional repository with meaningful commits, excellent README, clear	1 to >0 pts Good Mostly clear report with minor clarity or incomplete organisation issues; missing major good repository with minor missing elements or weak walkthrough.	0 pts Poor Poorly structured or incomplete report; submission requirements.	2 / 2 pts		

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3. Implementation and Testing				
Criteria	Ratings		Points	
	installation instructions, and a strong technical walkthrough video linked in README.			
				Total points: 10
Supervisor assessment of the students availability Assignment 5: Supervisor's grade [F:A]	20 Jul by 23:59	20 Jul at 22:09	 / 10	
Capstone Report Submission Assignment 4: Capstone Report Submission [F:A]	24 Jul by 23:59	24 Jul at 23:18	 / 35	
Mission Capstone Exit Ticket: Compiling a Mission portfolio Imported Assignments	27 Jul by 23:59	20 Jul at 14:25		
Defense [Presentation and Demo] Assignment 6: Capstone Defense [S:A]	31 Jul by 23:59		 / 30	
Final Submission Assignment 7: Final Capstone Report Submission [S:A]	31 Jul by 23:59		 / 5	
Roll Call Attendance Workshop Attendance [F:A]		29 May at 11:16	100%	
Assignments			100%	1.00 / 1.00

Name	Due	Submitted	Status	Score
Assignment 1: Research proposal [F:A]				82.11%15.60 / 19.00
Assignment 2: Initial Design and Development [F:A]				N/A0.00 / 0.00
Assignment 3: Final Capstone Implementation [S:A]				66.67%10.00 / 15.00
Assignment 4: Capstone Report Submission [F:A]				N/A0.00 / 0.00
Workshop Attendance [F:A]				100%100.00 / 100.00
Assignment 5: Supervisor's grade [F:A]				N/A0.00 / 0.00
Assignment 6: Capstone Defense [S:A]				N/A0.00 / 0.00
Assignment 7: Final Capstone Report Submission [S:A]				N/A0.00 / 0.00
Imported Assignments				N/A0.00 / 0.00
Total				78.46%

